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Paenibacillus strains and methods for their use

Abstract

The present invention provides a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the fusaricidin is produced with a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the tridecaptin is produced with a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7. Also provided are methods of treating an agricultural plant to control a disease and methods of increasing the vigor and/or crop yield of an agricultural plant with the disclosed compositions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a 35 U.S.C. § 371 national phase entry of International Application No. PCT/US2020/021125, filed on Mar. 5, 2020, which claims the benefit under 35 U.S.C. § 119 of U.S. Provisional Patent Application No. 62/815,069, filed Mar. 7, 2019, the entire contents of each of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates to the field of bacterial strains and their ability to control plant diseases and to increase plant vigor and crop yield. In particular, the present invention is directed to *Paenibacillus* sp. strains producing fusaricidins and tridecaptins and their use to maintain plant health and improve plant growth.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

(2) The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII-formatted sequence listing with a file named “BCS189006WO_ST25.txt” created on Feb. 25, 2020, and having a size of 185 kilobytes, and is filed concurrently with the specification. The sequence listing contained in this ASCII-formatted document is part of the specification and is herein incorporated by reference in its entirety.

BACKGROUND

(3) Fungicides have myriad uses, including for crop protection; as food, feed, and cosmetics preservatives; and as therapeutic agents for both human and veterinary applications. Crop yield reduction, foodborne diseases and fungal infections of both humans and animals are a problem in both developed and developing countries.

(4) Synthetic insecticides or fungicides often are non-specific and therefore can act on organisms other than the target ones, including other naturally occurring beneficial organisms. Because of their chemical nature, they may also be toxic and non-biodegradable. Consumers worldwide are increasingly conscious of the potential environmental and health problems associated with the residuals of chemicals, particularly in food products. This has resulted in growing consumer pressure to reduce the use or at least the quantity of chemical (i.e., synthetic) pesticides. Thus, there is a need to manage food chain requirements while still allowing effective pest control.

(5) A further problem arising with the use of synthetic insecticides or fungicides is that the repeated and exclusive application of an insecticide or fungicides often leads to selection of resistant pathogenic microorganisms. Normally, such strains are also cross-resistant against other active ingredients having the same mode of action. An effective control of the pathogens with said active compounds is then not possible any longer. However, active ingredients having new mechanisms of action are difficult and expensive to develop.

(6) The risk of resistance development in pathogen populations as well as environmental and human health concerns have fostered interest in identifying alternatives to synthetic insecticides and fungicides for managing plant diseases. The use of biological control agents is one alternative.

(7) *Paenibacillus* is a genus of low GC-content, endospore-forming, Gram-positive bacteria (Firmicutes). Bacteria belonging to this genus are prolific producers of industrially-relevant extracellular enzymes and antimicrobial substances, including non-ribosomal peptide classes. There is considerable variability between species and strains within the *Paenibacillus* genus with only certain strains showing efficacy in controlling plant pathogens and in promoting plant growth.

(8) There is a need for *Paenibacillus* sp. strains with enhanced fungicidal activity and the ability to improve plant growth and vigor. Improvements to the efficacy of existing fungicides and the development of alternatives that are not susceptible to development of fungal resistance are highly desirable.

SUMMARY

(9) The present invention is directed to a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7.

(10) In certain aspects, the fusaricidin synthetase gene of fusA is encoded by a DNA sequence comprising SEQ ID NO: 3 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence. In other aspects, the NRPS gene of triE is encoded by a DNA sequence comprising SEQ ID NO: 7 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

(11) In some embodiments, the composition comprises Fusaricidin A, Fusaricidin B, Fusaricidin C, Fusaricidin D, LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and/or LiF07b.

(12) In other embodiments, the composition effectively controls a plant disease caused by a fungus selected from the group consisting of *Botrytis cinerea*, *Sphaerotheca fuliginea*, and *Puccinia triticina*.

(13) In certain aspects, the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

(14) In other aspects, the composition comprises a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

(15) In one aspect, the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723,

or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.

(16) In some embodiments, the *Paenibacillus* sp. strain further comprises a nitrogen fixation gene cluster; and the nitrogen fixation gene cluster comprises a nitrogen fixation gene of *nifB* encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10.

(17) In one embodiment, the nitrogen fixation gene cluster comprises a nitrogen fixation gene of *nifB* encoded by a DNA sequence comprising SEQ ID NO: 10 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

(18) In other embodiments, the expression of the nitrogen fixation gene cluster contributes to enhanced plant growth, plant vigor, and/or crop yield. In yet other embodiments, the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of *fusA* encoded by a DNA sequence exhibiting at least 97.3% sequence identity to SEQ ID NO: 1 and the *Paenibacillus* sp. strain comprises a NRPS gene of *triE* encoded by a DNA sequence exhibiting at least 97.5% sequence identity to SEQ ID NO: 5.

(19) In one aspect, the fusaricidin synthetase gene of *fusA* is encoded by a DNA sequence comprising SEQ ID NO: 1 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence. In another aspect, the NRPS gene of *triE* is encoded by a DNA sequence comprising SEQ ID NO: 5 or a degenerate nucleotide sequence thereof encoding the same amino acid sequence.

(20) In certain aspects, the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

(21) In other aspects, the composition comprises a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

(22) In some embodiments, the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724.

(23) In certain aspects, the present invention provides a method of treating an agricultural plant to control a disease, wherein the method comprises applying an effective amount of a composition disclosed herein to the plant, to a part of the plant and/or to a locus of the plant.

(24) In one aspect, the method comprises applying the composition to foliar plant parts. In another aspect, the composition is applied at about 1×10^8 to about 1×10^{12} colony forming units (CFU) of the *Paenibacillus* sp. strain per hectare or at about 0.5 kg to about 5 kg fermentation solids per hectare. In yet another aspect, the disease is a fungal disease or a bacterial disease.

(25) In other embodiments, the present invention provides a method for increasing the vigor and/or crop yield of an agricultural plant, wherein the plant, the plant propagule, the seed of the plant and/or a locus where the plant is growing or is intended to grow is treated with an effective amount of a composition disclosed herein.

(26) In one embodiment, the treatment is carried out as an in-furrow treatment, seed treatment, and/or foliar treatment.

(27) In another embodiment, the agricultural plant is selected from the group consisting of soybean, corn, wheat, triticale, barley, oat, rye, rape, millet, rice, sunflower, cotton, sugar beet, pome fruit, stone fruit, citrus, banana, strawberry, blueberry, almond, grape, mango, papaya, peanut, potato, tomato, pepper, cucurbit, cucumber, melon, watermelon, garlic, onion, broccoli, carrot, cabbage, bean, dry bean, canola, pea, lentil, alfalfa, trefoil, clover, flax, elephant grass, grass, lettuce, sugarcane, tea, tobacco and coffee; each in its natural or genetically modified form.

(28) In yet another embodiment, the present invention relates to a seed treated with a composition disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts the control of *Sphaerotheca fuliginea* (PODOXA) achieved with *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 compared to several other *Paenibacillus* sp. strains.

(2) FIG. 2 depicts the plant growth promotion in tomato plants resulting from drench treatments with whole broth cultures of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.

DETAILED DESCRIPTION

(3) The microorganisms and particular strains described herein, unless specifically noted otherwise, are all separated from nature and grown under artificial conditions such as in shake flask cultures or through scaled-up manufacturing processes, such as in bioreactors to maximize bioactive metabolite production, for example. Growth under such conditions leads to strain “domestication.” Generally, such a “domesticated” strain differs from its counterparts found in nature in that it is cultured as a homogenous population that is not subject to the selection pressures found in the natural environment but rather to artificial selection pressures.

(4) As used herein, the verb “comprise” as is used in this description and in the claims and its conjugations are used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not

excluded. In addition, reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the elements are present, unless the context clearly requires that there is one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one”.

(5) In certain aspects, the *Paenibacillus* sp. strain of the present invention is selected from any one of the following: *P. terrae*, *P. brasilensis*, *P. polymyxa*, or *P. peoriae*. In one aspect, the *Paenibacillus* sp. strain of the present invention is *P. terrae*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. brasilensis*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. peoriae*. In another aspect, the *Paenibacillus* sp. strain of the present invention is *P. polymyxa*.

(6) In one embodiment, a mutant strain of the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 is provided. The term “mutant” refers to a genetic variant derived from *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. In one embodiment, the mutant has one or more or all the identifying (functional) characteristics of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. In a particular instance, the mutant or a fermentation product thereof controls (as an identifying functional characteristic) fungi, oomycetes and/or bacteria at least as well as the parent *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. Such mutants may be genetic variants having a genomic sequence that has greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724. Mutants may be obtained by treating cells of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 with chemicals or irradiation or by selecting spontaneous mutants from a population of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 cells (such as phage resistant or antibiotic resistant mutants), by genome shuffling, as described below, or by other means well known to those practiced in the art.

(7) Genome shuffling among *Paenibacillus* strains can be facilitated through the use of a process called protoplast fusion. The process begins with the formation of protoplasts from vegetative bacillary cells. The removal of peptidoglycan cell wall, typically using lysozyme and an osmotic stabilizer, results in the formation of a protoplast. This process is visible under a light microscope with the appearance of spherical cells. Addition of PEG, polyethylene glycol, then induces fusion among protoplasts, allowing genetic contents of two or more cells to come in contact facilitating recombination and genome shuffling. Fused cells then repartition and are recovered on a solid growth medium. During recovery, protoplasts rebuild peptidoglycan cell walls, transitioning back to bacillary shape. See Schaeffer, et. al., (1976) PNAS USA, vol. 73, 6:2151-2155).

(8) In certain aspects, the present invention is directed to *Paenibacillus* sp. strains producing fusaricidins. The fusaricidins are a family of depsipeptides with a 15-guanidino-3-hydroxypentadecanoic acid (GHPD) tail, as well as their linear counterparts. The specific conserved characteristics of fusaricidins are this GHPD tail, as well as three of the six amino acids in the sequence: (1) Threonine, (4) Threonine, and (6) Alanine.

(9) Originally discovered but not characterized by Nakajima et al. (*J. Antibiot.* 1972, 25, 243-247) in the mid-70's, fusaricidins were described by Kurusu et al. (*J. Antibiot.*, 1987, 40, 1506-1514) in the late 1980's. They were further studied by Kajimura et al. (*J. Antibiot.*, 1996, 49, 129-135; *J. Antibiot.*, 1997 50, 220-228), Kuroda et al. (*Heterocycles*, 2000, 53, 1533-1549; *J. Mass Spectrom.*, 2001, 36, 30-37), and Beatty et al. (*Can. J. Microbiol.*, 2002, 48, 159-169) throughout the mid-1990's to the early 2000's. During this period of heavy investigation these compounds were renamed several times depending on the author (Fusaricidin A is also known as LiF04a, Gatavalin, or even KT-6291A). Though there are many publications on the topic, select compounds from the same group of 24 known fusaricidins is described each time.

(10) After a somewhat quiet period on the topic, Vater et al. (*J. Am. Soc. Mass Spectrom.*, 2015, 26, 1130-1141) described the structural elucidation of fusaricidins by mass spectrometry and described several analogs of the family. Vater et al. identified a new class of fusaricidin-like compounds with seven amino acids (i.e., an extra alanine connected to the (4) threonine residue in the peptide sequence). As used herein, the term “acyclic analog” refers to the compound that corresponds to the fusaricidin or fusaricidin-like compound but lacks the ester bond, resulting in a linear structure.

(11) The amino acid chains of fusaricidins are linked together and modified by a non-ribosomal peptide synthetase (NRPS). The multi-domain NRPS consists of up to 15,000 amino acids and is therefore considered among the longest proteins in nature (Schwarzer et al., (2003) Nonribosomal Peptides: From Genes to Products. *Nat. Prod. Rep.* 20, 275-287). NRPS incorporation is not limited to the 21 standard amino acids translated by the ribosome, and this promiscuity contributes to the great structural diversity and biological activity of non-ribosomal peptides (Li and Jensen, (2008). Nonribosomal biosynthesis of fusaricidins by *Paenibacillus polymyxa* PKB1 involves direct activation of a d-amino acid. *Chem. Biol.* 15, 118-127).

(12) In *P. polymyxa* E68, the fusaricidin biosynthetic gene cluster (fusGFEDCBA) has been characterized, and the NRPS coding sequence, the largest coding DNA sequence (CDS) in the cluster, was observed to encode a six-module peptide (Choi et al., Identification and Functional Analysis of the Fusaricidin Biosynthetic Gene of *Paenibacillus polymyxa* E681. *Biochem. Biophys. Res. Commun.* 365, 89-95; Li and Jensen, Identification and Functional Analysis of the Fusaricidin Biosynthetic Gene of *Paenibacillus polymyxa* E681. *Biochem. Biophys. Res. Commun.* 365, 89-95; Li et al., (2013).

Promoter Analysis and Transcription Regulation of fus Gene Cluster Responsible for Fusaricidin Synthesis of *Paenibacillus polymyxa* SQR-21. Appl. Microbiol. Biotechnol. 97, 9479-9489). The biosynthetic cluster includes other CDS responsible for biosynthesis of the lipid moiety but does not contain transporter genes (Li and Jensen, (2008). Nonribosomal biosynthesis of fusaricidins by *Paenibacillus polymyxa* PKB1 involves direct activation of a d-amino acid. Chem. Biol. 15, 118-127). In *P. polymyxa*, a promoter for the fus operon was identified and shown to be bound by a transcriptional repressor (AbrB) which previous studies implicated as a regulator of sporulation; this is of interest since fusaricidin was observed to be synthesized during sporulation, thus coordinating the microbe's secondary metabolism with its life cycle (Li et al., (2013). Promoter Analysis and Transcription Regulation of fus Gene Cluster Responsible for Fusaricidin Synthesis of *Paenibacillus polymyxa* SQR-21. Appl. Microbiol. Biotechnol. 97, 9479-9489).

(13) In other aspects, the present invention is directed to *Paenibacillus* sp. strains producing tridecaptins. The tridecaptins are a group of linear cationic lipopeptides analogous to the polymyxins. The acyl tridecapeptides were first isolated in 1978 and display antimicrobial activity against Gram-positive and Gram-negative bacteria. Shoji, Junichi et al., Journal of Antibiotics (1978), 31 (7), 646-51. More recently, tridecaptin A.sub.1 has been characterized in a strain of *Paenibacillus terrae*. Lohans, Christopher T. et al., ChemBioChem (2014), 15 (2), 243-249.

(14) Allelic diversity is typically thought to be responsible for producing chemical diversity. However, an interesting feature of the fus cluster is that a diversity of fusaricidins, differing in their incorporated amino acids (Tyr, Val, Be, allo-Ile, Phe), can be produced by a single allele of fusA; the underlying mechanism is that the NRPS A-domain, responsible for recognition of amino acids, has relaxed substrate specificity (Han et al., (2012). Site-Directed Modification of the Adenylation Domain of the Fusaricidin Nonribosomal Peptide Synthetase for Enhanced Production of Fusaricidin Analogs. Biotechnol. Lett. 34, 1327-1334; Mousa et al., (2015) Biodiversity of Genes Encoding Anti-Microbial Traits within Plant Associated Microbes, Front Plant Sci. 2015; 6: 231).

(15) In certain embodiments, the composition comprises a biologically pure culture of a *Paenibacillus* sp. strain comprising a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3 or SEQ ID NO: 1, a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7 or SEQ ID NO: 5, and/or a nitrogen fixation gene of nifB encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10. In some aspects the sequence identity between the fusA, triE, or nifB of the *Paenibacillus* sp. strain and the respective sequence (i.e., SEQ ID NOs: 1 or 3 for fusA; SEQ ID NOs: 5 or 7 for triE; SEQ ID NO: 10 for nifB) is at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%. In other aspects, the sequence identity between the fusA, triE, or nifB of the *Paenibacillus* sp. strain and the respective sequence (i.e., SEQ ID NOs: 1 or 3 for fusA; SEQ ID NOs: 5 or 7 for triE; SEQ ID NO: 10 for nifB) is between about 80% and 100%, between about 85% and 100%, between about 90% and 100%, or between about 95% and 100%.

(16) Due to the degeneracy of the genetic code, different nucleotide codons may be used to code for a particular amino acid. A host cell often displays a preferred pattern of codon usage. Structural nucleic acid sequences are preferably constructed to utilize the codon usage pattern of the particular host cell. This generally enhances the expression of the structural nucleic acid sequence in a transformed host cell. Any of the nucleic acid sequences disclosed herein may be modified to reflect the preferred codon usage of a host cell or organism in which they are contained.

(17) Additional variations in the nucleic acid sequences described herein may encode proteins having equivalent or superior characteristics when compared to the proteins from which they are engineered. Mutations may include deletions, insertions, truncations, substitutions, fusions, shuffling of motif sequences, and the like.

(18) Mutations to a nucleic acid sequence may be introduced in either a specific or random manner, both of which are well known to those of skill in the art of molecular biology. A myriad of site-directed mutagenesis techniques exist, typically using oligonucleotides to introduce mutations at specific locations in a nucleic acid sequence. Examples include single strand rescue (Kunkel et al., Proc. Natl. Acad. Sci. U.S.A., 82: 488-492, 1985), unique site elimination (Deng and Nickloff, Anal. Biochem. 200:81, 1992), nick protection (Vandeyar, et al. Gene 65: 129-133, 1988), and PCR (Costa et al., Methods Mol. Biol. 57: 31-44, 1996). Random or non-specific mutations may be generated by chemical agents (for a general review, see Singer and Kusmierk, Ann. Rev. Biochem. 52: 655-693, 1982) such as nitrosoguanidine (Cerdeira-Olmedo et al., J. Mol. Biol. 33: 705-719, 1968; Guerola, et al. Nature New Biol. 230: 122-125, 1971) and 2-aminopurine (Rogan and Bessman, J. Bacteriol. 103: 622-633, 1970); or by biological methods such as passage through mutator strains (Greener, et al. Mol. Biotechnol. 7: 189-195, 1997).

(19) The modifications may result in either conservative or non-conservative changes in the amino acid sequence. Conservative changes result from additions, deletions, substitutions, etc. in the structural nucleic acid sequence which do not alter the final amino acid sequence of the protein. In a preferred embodiment, the encoded protein has between 20 and 500 conservative changes, more preferably between 15 and 300 conservative changes, even more preferably between 10 and 150 conservative changes, and most preferably between 5 and 75 conservative changes.

(20) Non-conservative changes include additions, deletions, and substitutions which result in an altered amino acid sequence. In a preferred embodiment, the encoded protein has between 10 and 250 non-conservative amino acid changes, more preferably between 5 and 100 non-conservative amino acid changes, even more preferably between 2 and 50 non-conservative amino acid changes, and most preferably between 1 and 30 non-conservative amino acid changes.

(21) Additional methods of making the alterations described above are described by Ausubel et al., Current Protocols in Molecular Biology, John Wiley and Sons, Inc., 1995, Bauer et al., Gene, 37:73, 1985; Craik, BioTechniques, 3: 12-19, 1985; Frits Eckstein et al., Nucleic Acids Research, 10: 6487-6497, 1982; Sambrook, et al., Molecular Cloning: A

Laboratory Manual, Second Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1989, Smith, et al., In: Genetic Engineering: Principles and Methods, Setlow et al., Eds., Plenum Press, N.Y., 1-32, 1981, and Osuna, et al., Critical Reviews In Microbiology, 20: 107-116, 1994.

(22) Modifications may be made to the protein sequences of the present invention and the nucleic acid segments which encode them that maintain the desired properties of the molecule. The following is a discussion based upon changing the amino acid sequence of a protein to create an equivalent, or possibly an improved molecule.

(23) Certain amino acids may be substituted for other amino acids in a protein sequence without appreciable loss of the desired activity.

(24) It is thus contemplated that various changes may be made in the peptide sequences of the disclosed protein sequences, or their corresponding nucleic acid sequences without appreciable loss of the biological activity.

(25) In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biological function on a protein is generally understood in the art (Kyte and Doolittle, *J. Mol. Biol.*, 157: 105-132, 1982). It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like.

(26) Each amino acid has been assigned a hydropathic index on the basis of their hydrophobicity and charge characteristics. These are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cysteine (+2.5); methionine (+1.9); alanine (+1.8); glycine (−0.4); threonine (−0.7); serine (−0.8); tryptophan (−0.9); tyrosine (−1.3); proline (−1.6); histidine (−3.2); glutamate/glutamine/aspartate/asparagine (−3.5); lysine (−3.9); and arginine (−4.5).

(27) It is known in the art that certain amino acids may be substituted by other amino acids having a similar hydropathic index or score and still result in a protein with similar biological activity, i.e., still obtain a biologically functional protein. In making such changes, the substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are more preferred, and those within ± 0.5 are most preferred.

(28) It is also understood in the art that the substitution of like amino acids may be made effectively on the basis of hydrophilicity. U.S. Pat. No. 4,554,101 (Hopp) states that the greatest local average hydrophilicity of a protein, as governed by the hydrophilicity of its adjacent amino acids, correlates with a biological property of the protein. The following hydrophilicity values have been assigned to amino acids: arginine/lysine (+3.0); aspartate/glutamate (+3.0 $\pm$ 1); serine (+0.3); asparagine/glutamine (+0.2); glycine (0); threonine (−0.4); proline (−0.5 $\pm$ 1); alanine/histidine (−0.5); cysteine (−1.0); methionine (−1.3); valine (−1.5); leucine/isoleucine (−1.8); tyrosine (−2.3); phenyl alanine (−2.5); and tryptophan (−3.4).

(29) It is understood that an amino acid may be substituted by another amino acid having a similar hydrophilicity score and still result in a protein with similar biological activity, i.e., still obtain a biologically functional protein. In making such changes, the substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are more preferred, and those within ± 0.5 are most preferred.

(30) As outlined above, amino acid substitutions are therefore based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions which take various of the foregoing characteristics into consideration are well known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine, and isoleucine.

(31) The present invention also encompasses methods of treating a plant to control plant diseases by administering to a plant or a plant part, such as a leaf, stem, flowers, fruit, root, or seed or by applying to a locus on which plant or plant parts grow, such as soil, the disclosed *Paenibacillus* sp. strains or mutants thereof, or cell-free preparations thereof or metabolites thereof.

(32) In a method according to the invention a composition containing a disclosed *Paenibacillus* sp. strain or a fungicidal mutant thereof can be applied to any plant or any part of any plant grown in any type of media used to grow plants (e.g., soil, vermiculite, shredded cardboard, and water) or applied to plants or the parts of plants grown aurally, such as orchids or staghorn ferns. The composition may for instance be applied by spraying, atomizing, vaporizing, scattering, dusting, watering, squirting, sprinkling, pouring or fumigating. As already indicated above, application may be carried out at any desired location where the plant of interest is positioned, such as agricultural, horticultural, forest, plantation, orchard, nursery, organically grown crops, turfgrass and urban environments.

(33) Compositions of the present invention can be obtained by culturing the disclosed *Paenibacillus* sp. strains or a fungicidal mutant (strain) derived therefrom according to methods well known in the art, including by using the media and other methods described in the examples below. Conventional large-scale microbial culture processes include submerged fermentation, solid state fermentation, or liquid surface culture. Towards the end of fermentation, as nutrients are depleted, cells begin the transition from growth phase to sporulation phase, such that the final product of fermentation is largely spores, metabolites and residual fermentation medium. Sporulation is part of the natural life cycle of *Paenibacillus* and is generally initiated by the cell in response to nutrient limitation. Fermentation is configured to obtain high levels of colony forming units of and to promote sporulation. The bacterial cells, spores and metabolites in culture media resulting from fermentation may be used directly or concentrated by conventional industrial methods, such as centrifugation, tangential-flow filtration, depth filtration, and evaporation.

(34) Compositions of the present invention include fermentation products. In some embodiments, the concentrated fermentation broth is washed, for example, via a diafiltration process, to remove residual fermentation broth and metabolites. The term “broth concentrate,” as used herein, refers to whole broth (fermentation broth) that has been

concentrated by conventional industrial methods, as described above, but remains in liquid form. The term “fermentation solid,” as used herein, refers to the solid material that remains after the fermentation broth is dried. The term “fermentation product,” as used herein, refers to whole broth, broth concentrate and/or fermentation solids. Compositions of the present invention include fermentation products.

(35) The fermentation broth or broth concentrate can be dried with or without the addition of carriers using conventional drying processes or methods such as spray drying, freeze drying, tray drying, fluidized-bed drying, drum drying, or evaporation.

(36) The resulting dry products may be further processed, such as by milling or granulation, to achieve a specific particle size or physical format. Carriers, described below, may also be added post-drying.

(37) Cell-free preparations of fermentation broth of the strains of the present invention can be obtained by any means known in the art, such as extraction, centrifugation and/or filtration of fermentation broth. Those of skill in the art will appreciate that so-called cell-free preparations may not be devoid of cells but rather are largely cell-free or essentially cell-free, depending on the technique used (e.g., speed of centrifugation) to remove the cells. The resulting cell-free preparation may be dried and/or formulated with components that aid in its application to plants or to plant growth media.

Concentration methods and drying techniques described above for fermentation broth are also applicable to cell-free preparations.

(38) In one embodiment, the fermentation product comprises at least about $1 \times 10^{4.4}$ colony forming units (CFU) of the microorganism (e.g., *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 or a fungicidal mutant strain thereof)/mL broth. In another embodiment, the fermentation product comprises at least about $1 \times 10^{5.5}$ colony forming units (CFU) of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about $1 \times 10^{6.6}$ CFU of the microorganism/mL broth. In yet another embodiment, the fermentation product comprises at least about $1 \times 10^{7.7}$ CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about $1 \times 10^{8.8}$ CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about $1 \times 10^{9.9}$ CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^{10} CFU of the microorganism/mL broth. In another embodiment, the fermentation product comprises at least about 1×10^{11} CFU of the microorganism/mL broth.

(39) The inventive compositions can be used as such or, depending on their particular physical and/or chemical properties, in the form of their formulations or the use forms prepared therefrom, such as aerosols, capsule suspensions, cold-fogging concentrates, warm-fogging concentrates, encapsulated granules, fine granules, flowable concentrates for the treatment of seed, ready-to-use solutions, dustable powders, emulsifiable concentrates, oil-in-water emulsions, water-in-oil emulsions, macrogranules, microgranules, oil-dispersible powders, oil-miscible flowable concentrates, oil-miscible liquids, gas (under pressure), gas generating product, foams, pastes, pesticide coated seed, suspension concentrates, oil dispersion, suspo-emulsion concentrates, soluble concentrates, suspensions, wettable powders, soluble powders, dusts and granules, water-soluble and water-dispersible granules or tablets, water-soluble and water-dispersible powders for the treatment of seed, wettable powders, natural products and synthetic substances impregnated with active ingredient, and also microencapsulations in polymeric substances and in coating materials for seed, and also ULV cold-fogging and warm-fogging formulations.

(40) In some embodiments, the inventive compositions are liquid formulations. Non-limiting examples of liquid formulations include suspension concentrations and oil dispersions. In other embodiments, the inventive compositions are solid formulations. Non-limiting examples of liquid formulations include freeze-dried powders and spray-dried powders.

(41) All plants and plant parts can be treated in accordance with the invention. In the present context, plants are understood as meaning all plants and plant populations, such as desired and undesired wild plants or crop plants (including naturally occurring crop plants). Crop plants can be plants which can be obtained by traditional breeding and optimization methods or by biotechnological and recombinant methods, or combinations of these methods, including the transgenic plants and including the plant varieties capable or not of being protected by Plant Breeders' Rights. Plant parts are understood as meaning all aerial and subterranean parts and organs of the plants, such as shoot, leaf, flower and root, examples which may be mentioned being leaves, needles, stalks, stems, flowers, fruiting bodies, fruits and seeds, and also roots, tubers and rhizomes. The plant parts also include crop material and vegetative and generative propagation material, for example cuttings, tubers, rhizomes, slips and seeds.

(42) As has already been mentioned above, all plants and their parts may be treated in accordance with the invention. In a preferred embodiment, plant species and plant varieties, and their parts, which grow wild or which are obtained by traditional biological breeding methods such as hybridization or protoplast fusion are treated. In a further preferred embodiment, transgenic plants and plant varieties which have been obtained by recombinant methods, if appropriate in combination with traditional methods (genetically modified organisms), and their parts are treated. The term “parts” or “parts of plants” or “plant parts” has been explained hereinabove. Plants of the plant varieties which are in each case commercially available or in use are especially preferably treated in accordance with the invention. Plant varieties are understood as meaning plants with novel traits which have been bred both by traditional breeding, by mutagenesis or by recombinant DNA techniques. They may take the form of varieties, races, biotypes and genotypes.

(43) The treatment of the plants and plant parts with the compositions according to the invention is carried out directly or by acting on the environment, habitat or storage space using customary treatment methods, for example by dipping, spraying, atomizing, misting, evaporating, dusting, fogging, scattering, foaming, painting on, spreading, injecting, drenching, trickle

irrigation and, in the case of propagation material, in particular in the case of seed, furthermore by the dry seed treatment method, the wet seed treatment method, the slurry treatment method, by encrusting, by coating with one or more coats and the like. It is furthermore possible to apply the active substances by the ultra-low volume method or to inject the active substance preparation or the active substance itself into the soil.

(44) A preferred direct treatment of the plants is the leaf application treatment, i.e., compositions according to the invention are applied to the foliage, it being possible for the treatment frequency and the application rate to be matched to the infection pressure of the pathogen in question.

(45) Preferred plants are those from the group of the useful plants, ornamentals, turfs, generally used trees which are employed as ornamentals in the public and domestic sectors, and forestry trees. Forestry trees comprise trees for the production of timber, cellulose, paper and products made from parts of the trees.

(46) The term “agricultural plants” as used in the present context refers to crop plants which are employed as plants for obtaining foodstuffs, feedstuffs, fuels or for industrial purposes.

(47) The terms “plant propagation material” and “plant propagule” are to be understood to denote all the generative parts of the plant such as seeds and vegetative plant material such as cuttings and tubers (e.g., potatoes), which can be used for the multiplication of the plant. This includes seeds, roots, fruits, tubers, bulbs, rhizomes, shoots, sprouts and other parts of plants. Seedlings and young plants, which are to be transplanted after germination or after emergence from soil, may also be mentioned. These young plants may also be treated totally or partially by immersion or pouring before transplantation.

(48) The term “locus” is to be understood as any type of environment, soil, area or material where the plant is growing or intended to grow as well as the environmental conditions (such as temperature, water availability, radiation) that have an influence on the growth and development of the plant and/or its propagules. In addition, the term “locus” is to be understood as a plant, seed, soil, area, material or environment in which a pest is growing or may grow.

(49) “Crop yield” is an indicator for the condition of the plant, whereas “crop” is to be understood as any plant or plant product which is further utilized after harvesting, e.g., fruits in the proper sense, vegetables, nuts, grains, seeds, wood (e.g., in the case of silviculture plants), flowers (e.g., in the case of gardening plants, ornamentals) etc., that is anything of economic value that is produced by the plant.

(50) According to the present invention, “increased yield” of a plant, in particular of an agricultural, silvicultural and/or ornamental plant means that the yield of a product of the respective plant is increased by a measurable amount over the yield of the same product of the plant produced under the same conditions, but without the application of the composition of the invention.

(51) The term “seed” embraces seeds and plant propagules of all kinds including but not limited to true seeds, seed pieces, suckers, corms, bulbs, fruit, tubers, grains, cuttings, cut shoots and the like and means in a preferred embodiment true seeds.

(52) The term “seed treatment” comprises all suitable seed treatment techniques known in the art, such as seed dressing, seed coating, seed dusting, seed soaking, seed impregnation and seed pelleting.

(53) The agricultural plants which can be treated and/or improved with the compositions and methods of the present invention include for example the following types of plants: turf, vines, cereals, for example wheat, barley, rye, oats, rice, maize and millet/sorghum; beet, for example sugar beet and fodder beet; fruits, for example pome fruit, stone fruit and soft fruit, for example apples, pears, plums, peaches, almonds, cherries and berries, for example strawberries, raspberries, blackberries; legumes, for example beans, lentils, peas and soybeans; oil crops, for example oilseed rape, mustard, poppies, olives, sunflowers, coconuts, castor oil plants, cacao and peanuts; cucurbits, for example pumpkin/squash, cucumbers and melons; fibre plants, for example cotton, flax, hemp and jute; citrus fruit, for example oranges, lemons, grapefruit and tangerines; vegetables, for example spinach, lettuce, asparagus, cabbage species, carrots, onions, tomatoes, potatoes and bell peppers; *Lauraceae*, for example avocado, *Cinnamomum*, camphor, or else plants such as tobacco, nuts, coffee, aubergine, sugar cane, tea, pepper, grapevines, hops, bananas, latex plants and ornamentals, for example flowers, shrubs, deciduous trees and coniferous trees. This enumeration is no limitation.

(54) The following plants are considered to be particularly suitable target crops for applying compositions and methods of the present invention: cotton, aubergine, turf, pome fruit, stone fruit, soft fruit, maize, wheat, barley, cucumber, tobacco, vines, rice, cereals, pear, beans, soybeans, oilseed rape, tomato, bell pepper, melons, cabbage, potato and apple.

(55) Additional agricultural plants of particular interest include for example, cereals, for example wheat, rye, barley, triticale, oats or rice; beet, for example sugar beet or fodder beet; fruits, such as pomes, stone fruits or soft fruits, for example apples, pears, plums, peaches, almonds, cherries, strawberries, raspberries, blackberries or gooseberries; leguminous plants, such as lentils, peas, alfalfa or soybeans; oil plants, such as rape, mustard, olives, sunflowers, coconut, cocoa beans, castor oil plants, oil palms, ground nuts or soybeans; cucurbits, such as squashes, cucumber or melons; fiber plants, such as cotton, flax, hemp or jute; citrus fruit, such as oranges, lemons, grapefruits or mandarins; vegetables, such as broccoli, spinach, lettuce, asparagus, cabbages, carrots, onions, tomatoes, potatoes, cucurbits or paprika; lauraceous plants, such as avocados, cinnamon or camphor; energy and raw material plants, such as corn, soybean, rape, sugar cane or oil palm; corn; tobacco; nuts; coffee; tea; bananas; vines (table grapes and grape juice grape vines); hop; turf; natural rubber plants or ornamental and forestry plants, such as flowers, shrubs, broad-leaved trees or evergreens, for example conifers; and on the plant propagation material, such as seeds, and the crop material of these plants.

(56) Examples of trees which can be improved in accordance with the method according to the invention are: *Abies* sp., *Eucalyptus* sp., *Picea* sp., *Pinus* sp., *Aesculus* sp., *Platanus* sp., *Tilia* sp., *Acer* sp., *Tsuga* sp., *Fraxinus* sp., *Sorbus* sp., *Betula* sp., *Crataegus* sp., *Ulmus* sp., *Quercus* sp., *Fagus* sp., *Salix* sp., *Populus* sp.

(57) Preferred trees which can be improved in accordance with the method according to the invention are: from the tree

species *A. pariflora*, *A. carnea*; from the tree species *Platanus*: *P. aceriflora*, *P. occidentalis*, *P. racemosa*; from the tree species *Picea*: *P. abies*; from the tree species *Pinus*: *P. radiata*, *P. ponderosa*, *P. contorta*, *P. sylvestre*, *P. elliotii*, *P. montecola*, *P. albicaulis*, *P. resinosa*, *P. palustris*, *P. taeda*, *P. flexilis*, *P. jeffregi*, *P. baksiana*, *P. strobus*; from the tree species *Eucalyptus*: *E. grandis*, *E. globulus*, *E. camadensis*, *E. nitens*, *E. obliqua*, *E. regnans*, *E. pilularis*.

(58) Especially preferred trees which can be improved in accordance with the method according to the invention are: from the tree species *Pinus*: *P. radiata*, *P. ponderosa*, *P. contorta*, *P. sylvestre*, *P. strobus*; from the tree species *Eucalyptus*: *E. grandis*, *E. globulus*, *E. camadensis*.

(59) Very particularly preferred trees which can be improved in accordance with the method according to the invention are: horse chestnut, Platanaceae, linden tree, maple tree.

(60) The present invention can also be applied to any turf grasses, including cool-season turf grasses and warm-season turf grasses. Examples of cold-season turf grasses are bluegrasses (*Poa* spp.), such as Kentucky bluegrass (*Poa pratensis* L.), rough bluegrass (*Poa trivialis* L.), Canada bluegrass (*Poa compressa* L.), annual bluegrass (*Poa annua* L.), upland bluegrass (*Poa glauca* Gaudin), wood bluegrass (*Poa nemoralis* L.) and bulbous bluegrass (*Poa bulbosa* L.); bentgrasses (*Agrostis* spp.) such as creeping bentgrass (*Agrostis palustris* Huds.), colonial bentgrass (*Agrostis tenuis* Sibth.), velvet bentgrass (*Agrostis canina* L.), South German mixed bentgrass (*Agrostis* spp. including *Agrostis tenuis* Sibth., *Agrostis canina* L., and *Agrostis palustris* Huds.), and redtop (*Agrostis alba* L.); fescues (*Festuca* spp.), such as red fescue (*Festuca rubra* L. spp. *rubra*), creeping fescue (*Festuca rubra* L.), chewings fescue (*Festuca rubra commutata* Gaud.), sheep fescue (*Festuca ovina* L.), hard fescue (*Festuca longifolia* Thuill.), hair fescue (*Festuca capillata* Lam.), tall fescue (*Festuca arundinacea* Schreb.) and meadow fescue (*Festuca elanor* L.); ryegrasses (*Lolium* spp.), such as annual ryegrass (*Lolium multiflorum* Lam.), perennial ryegrass (*Lolium perenne* L.) and Italian ryegrass (*Lolium multiflorum* Lam.); and wheatgrasses (*Agropyron* spp.), such as fairway wheatgrass (*Agropyron cristatum* (L.) Gaertn.), crested wheatgrass (*Agropyron desertorum* (Fisch.) Schult.) and western wheatgrass (*Agropyron smithii* Rydb.)

(61) Examples of further cool-season turf grasses are beachgrass (*Ammophila breviligulata* Fern.), smooth brome grass (*Bromus inermis* Leyss.), cattails such as timothy (*Phleum pratense* L.), sand cattail (*Phleum subulatum* L.), orchardgrass (*Dactylis glomerata* L.), weeping alkaligrass (*Puccinellia distans* (L.) Parl.) and crested dog's-tail (*Cynosurus cristatus* L.)

(62) Examples of warm-season turf grasses are Bermuda grass (*Cynodon* spp. L. C. Rich), zoysia grass (*Zoysia* spp. Willd.), St. Augustine grass (*Stenotaphrum secundatum* Walt Kuntze), centipede grass (*Eremochloa ophiuroides* Munro Hack.), carpetgrass (*Axonopus affinis* Chase), Bahia grass (*Paspalum notatum* Flugge), Kikuyu grass (*Pennisetum clandestinum* Hochst. ex Chiov.), buffalo grass (*Buchloe dactyloids* (Nutt.) Engelm.), blue grama (*Bouteloua gracilis* (H.B.K.) Lag. ex Griffiths), seashore *paspalum* (*Paspalum vaginatum* Swartz) and sideoats grama (*Bouteloua curtipendula* (Michx. Torr.) Cool-season turf grasses are generally preferred for the use according to the invention. Especially preferred are bluegrass, benchgrass and redtop, fescues and ryegrasses. Bentgrass is especially preferred.

(63) The inventive compositions have potent microbicidal activity and can be used for control of unwanted microorganisms, such as fungi and bacteria, in crop protection and in the protection of materials.

(64) The invention also relates to a method for controlling unwanted microorganisms, characterized in that the inventive compositions are applied to the phytopathogenic fungi, phytopathogenic bacteria and/or their habitat.

(65) Fungicides can be used in crop protection for control of phytopathogenic fungi. They are characterized by an outstanding efficacy against a broad spectrum of phytopathogenic fungi, including soilborne pathogens, which are in particular members of the classes Plasmodiophoromycetes, Peronosporomycetes (Syn. Oomycetes), Chytridiomycetes, Zygomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes (Syn. Fungi imperfecti). Some fungicides are systemically active and can be used in plant protection as foliar, seed dressing or soil fungicide. Furthermore, they are suitable for combating fungi, which inter alia infest wood or roots of plant.

(66) Bactericides can be used in crop protection for control of Pseudomonadaceae, Rhizobiaceae, Enterobacteriaceae, Corynebacteriaceae and Streptomycetaceae.

(67) Non-limiting examples of pathogens of fungal diseases which can be treated in accordance with the invention include: diseases caused by powdery mildew pathogens, for example *Blumeria* species, for example *Blumeria graminis*;

Podosphaera species, for example *Podosphaera leucotricha*; *Sphaerotheca* species, for example *Sphaerotheca fuliginea*;

Uncinula species, for example *Uncinula necator*; diseases caused by rust disease pathogens, for example

Gymnosporangium species, for example *Gymnosporangium sabinae*; *Hemileia* species, for example *Hemileia vastatrix*;

Phakopsora species, for example *Phakopsora pachyrhizi* and *Phakopsora meibomiae*; *Puccinia* species, for example

Puccinia recondite, *P. triticea*, *P. graminis* or *P. striiformis*; *Uromyces* species, for example *Uromyces appendiculatus*;

diseases caused by pathogens from the group of the Oomycetes, for example *Albugo* species, for example *Albugo candida*;

Bremia species, for example *Bremia lactucae*; *Peronospora* species, for example *Peronospora pisi* or *P. brassicae*;

Phytophthora species, for example *Phytophthora infestans*; *Plasmopara* species, for example *Plasmopara viticola*;

Pseudoperonospora species, for example *Pseudoperonospora humuli* or *Pseudoperonospora cubensis*; *Pythium* species, for

example *Pythium ultimum*; leaf blotch diseases and leaf wilt diseases caused, for example, by *Alternaria* species, for

example *Alternaria solani*; *Cercospora* species, for example *Cercospora beticola*; *Cladosporium* species, for example

Cladosporium cucumerinum; *Cochliobolus* species, for example *Cochliobolus sativus* (conidia form: *Drechslera*, Syn:

Helminthosporium), *Cochliobolus miyabeanus*; *Colletotrichum* species, for example *Colletotrichum lindemuthianum*;

Cyloconium species, for example *Cyloconium oleaginum*; *Diaporthe* species, for example *Diaporthe citri*; *Elsinoe*

species, for example *Elsinoe fawcettii*; *Gloeosporium* species, for example *Gloeosporium laeticolor*; *Glomerella* species,

for example *Glomerella cingulata*; *Guignardia* species, for example *Guignardia bidwellii*; *Leptosphaeria* species, for example *Leptosphaeria maculans*, *Leptosphaeria nodorum*; *Magnaporthe* species, for example *Magnaporthe grisea*; *Marssonina* species, for example *Marssonina coronaria*; *Microdochium* species, for example *Microdochium nivale*; *Mycosphaerella* species, for example *Mycosphaerella graminicola*, *M. arachidicola* and *M. fijiensis*; *Phaeosphaeria* species, for example *Phaeosphaeria nodorum*; *Pyrenophora* species, for example *Pyrenophora teres*, *Pyrenophora tritici repentis*; *Ramularia* species, for example *Ramularia collo-cygni*, *Ramularia areola*; *Rhynchosporium* species, for example *Rhynchosporium secalis*; *Septoria* species, for example *Septoria apii*, *Septoria lycopersii*; *Typhula* species, for example *Typhula incarnata*; *Venturia* species, for example *Venturia inaequalis*; root and stem diseases caused, for example, by *Corticium* species, for example *Corticium graminearum*; *Fusarium* species, for example *Fusarium oxysporum*; *Gaeumannomyces* species, for example *Gaeumannomyces graminis*; *Rhizoctonia* species, such as, for example *Rhizoctonia solani*; *Sarocladium* diseases caused for example by *Sarocladium oryzae*; *Sclerotium* diseases caused for example by *Sclerotium oryzae*; *Tapesia* species, for example *Tapesia acuformis*; *Thielaviopsis* species, for example *Thielaviopsis basicola*; ear and panicle diseases (including corn cobs) caused, for example, by *Alternaria* species, for example *Alternaria* spp.; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium cladosporioides*; *Claviceps* species, for example *Claviceps purpurea*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Monographella* species, for example *Monographella nivalis*; *Septoria* species, for example *Septoria nodorum*; diseases caused by smut fungi, for example *Sphacelotheca* species, for example *Sphacelotheca reiliana*; *Tilletia* species, for example *Tilletia caries*, *T. controversa*; *Urocystis* species, for example *Urocystis occulta*; *Ustilago* species, for example *Ustilago nuda*, *U. nuda tritici*; fruit rot caused, for example, by *Aspergillus* species, for example *Aspergillus flavus*; *Botrytis* species, for example *Botrytis cinerea*; *Penicillium* species, for example *Penicillium expansum* and *P. purpurogenum*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Verticillium* species, for example *Verticillium albo-atrum*; seed and soilborne decay, mould, wilt, rot and damping-off diseases caused, for example, by *Alternaria* species, caused for example by *Alternaria brassicicola*; *Aphanomyces* species, caused for example by *Aphanomyces euteiches*; *Ascochyta* species, caused for example by *Ascochyta lentis*; *Aspergillus* species, caused for example by *Aspergillus flavus*; *Cladosporium* species, caused for example by *Cladosporium herbarum*; *Cochliobolus* species, caused for example by *Cochliobolus sativus*; (conidia form: *Drechslera*, *Bipolaris* Syn: *Helminthosporium*); *Colletotrichum* species, caused for example by *Colletotrichum coccodes*; *Fusarium* species, caused for example by *Fusarium culmorum*; *Gibberella* species, caused for example by *Gibberella zeae*; *Macrophomina* species, caused for example by *Macrophomina phaseolina*; *Monographella* species, caused for example by *Monographella nivalis*; *Penicillium* species, caused for example by *Penicillium expansum*; *Phoma* species, caused for example by *Phoma lingam*; *Phomopsis* species, caused for example by *Phomopsis sojae*; *Phytophthora* species, caused for example by *Phytophthora cactorum*; *Pyrenophora* species, caused for example by *Pyrenophora graminea*; *Pyricularia* species, caused for example by *Pyricularia oryzae*; *Pythium* species, caused for example by *Pythium ultimum*; *Rhizoctonia* species, caused for example by *Rhizoctonia solani*; *Rhizopus* species, caused for example by *Rhizopus oryzae*; *Sclerotium* species, caused for example by *Sclerotium rolfsii*; *Septoria* species, caused for example by *Septoria nodorum*; *Typhula* species, caused for example by *Typhula incarnata*; *Verticillium* species, caused for example by *Verticillium dahliae*; cancers, galls and witches' broom caused, for example, by *Nectria* species, for example *Nectria galligena*; wilt diseases caused, for example, by *Monilinia* species, for example *Monilinia laxa*; leaf blister or leaf curl diseases caused, for example, by *Exobasidium* species, for example *Exobasidium vexans*; *Taphrina* species, for example *Taphrina deformans*; decline diseases of wooden plants caused, for example, by Esca disease, caused for example by *Phaemoniella clamydosporea*, *Phaeoacremonium aleophilum* and *Fomitiporia mediterranea*; *Eutypa* *dyebae*, caused for example by *Eutypa lata*; *Ganoderma* diseases caused for example by *Ganoderma boninense*; *Rigidoporus* diseases caused for example by *Rigidoporus lignosus*; diseases of flowers and seeds caused, for example, by *Botrytis* species, for example *Botrytis cinerea*; diseases of plant tubers caused, for example, by *Rhizoctonia* species, for example *Rhizoctonia solani*; *Helminthosporium* species, for example *Helminthosporium solani*; Club root caused, for example, by *Plasmodiophora* species, for example *Plasmodiophora brassicae*; diseases caused by bacterial pathogens, for example *Xanthomonas* species, for example *Xanthomonas campestris* pv. *oryzae*; *Pseudomonas* species, for example *Pseudomonas syringae* pv. *lachrymans*; *Erwinia* species, for example *Erwinia amylovora*.

(68) The following diseases of soya beans can be controlled with preference:

(69) Fungal diseases on leaves, stems, pods and seeds caused, for example, by *Alternaria* leaf spot (*Alternaria* spec. *atrans tenuissima*), Anthracnose (*Colletotrichum gloeosporioides dematium* var. *truncatum*), brown spot (*Septoria glycines*), cercospora leaf spot and blight (*Cercospora kikuchii*), choanephora leaf blight (*Choanephora infundibulifera trispora* (Syn.)), *dactuliophora* leaf spot (*Dactuliophora glycines*), downy mildew (*Peronospora manshurica*), *drechslera* blight (*Drechslera glycini*), frog-eye leaf spot (*Cercospora sojae*), *leptosphaerulina* leaf spot (*Leptosphaerulina trifolii*), *phyllosticta* leaf spot (*Phyllosticta sojaecola*), pod and stem blight (*Phomopsis sojae*), powdery mildew (*Microsphaera diffusa*), *pyrenochaeta* leaf spot (*Pyrenochaeta glycines*), *rhizoctonia* aerial, foliage, and web blight (*Rhizoctonia solani*), rust (*Phakopsora pachyrhizi*, *Phakopsora meibomia*), scab (*Sphaceloma glycines*), *stemphylium* leaf blight (*Stemphylium botryosum*), target spot (*Corynespora cassiicola*).

(70) Fungal diseases on roots and the stem base caused, for example, by black root rot (*Calonectria crotalariae*), charcoal rot (*Macrophomina phaseolina*), *fusarium* blight or wilt, root rot, and pod and collar rot (*Fusarium oxysporum*, *Fusarium orthoceras*, *Fusarium semitectum*, *Fusarium equiseti*), *mycoleptodiscus* root rot (*Mycoleptodiscus terrestris*), *neocosmospora* (*Neocosmospora vasinfecta*), pod and stem blight (*Diaporthe phaseolorum*), stem canker (*Diaporthe*

phaseolora var. *calivora*), *Phytophthora megasperma*), brown stem rot (*Phialophora gregata*), *pythium* rot (*Pythium aphanidermatum*, *Pythium irregulare*, *Pythium debaryanum*, *Pythium myriotylum*, *Pythium ultimum*), *rhizoctonia* root rot, stem decay, and damping-off (*Rhizoctonia solani*), *sclerotinia* stem decay (*Sclerotinia sclerotiorum*), *sclerotinia* southern blight (*Sclerotinia rolsii*), *thielaviopsis* root rot (*Thielaviopsis basicola*).

(71) The inventive fungicidal compositions can be used for curative or protective/preventive control of phytopathogenic fungi. The invention therefore also relates to curative and protective methods for controlling phytopathogenic fungi by the use of the inventive compositions, which are applied to the seed, the plant or plant parts, the fruit or the soil in which the plants grow.

(72) The fact that the compositions are well tolerated by plants at the concentrations required for controlling plant diseases allows the treatment of above-ground parts of plants, of propagation stock and seeds, and of the soil.

(73) According to the invention all plants and plant parts can be treated including cultivars and plant varieties (whether or not protectable by plant variety or plant breeder's rights). Cultivars and plant varieties can be plants obtained by conventional propagation and breeding methods which can be assisted or supplemented by one or more biotechnological methods such as by use of double haploids, protoplast fusion, random and directed mutagenesis, molecular or genetic markers or by bioengineering and genetic engineering methods.

(74) In some embodiments, the compositions disclosed herein are used to enhance plant growth, plant vigor, and/or crop yield. "Yield" is to be understood as any plant product of economic value that is produced by the plant such as grains, fruits in the proper sense, vegetables, nuts, grains, seeds, wood (e.g., in the case of silviculture plants) or even flowers (e.g., in the case of gardening plants, ornamentals). The plant products may in addition be further utilized and/or processed after harvesting.

(75) According to the present invention, "increased yield" of a plant, in particular of an agricultural, silvicultural and/or ornamental plant means that the yield of a product of the respective plant is increased by a measurable amount over the yield of the same product of the plant produced under the same conditions, but without the application of the composition of the invention. Increased yield can be characterized, among others, by following improved properties of the plant: increased plant weight, increased plant height, increased biomass such as higher fresh and/or dry weight, higher grain yield, more tillers, larger leaves, increased shoot growth, increased protein content, increased oil content, increased starch content, increased pigment content

(76) According to one embodiment of the present invention, the yield is increased by at least 5%. According to another embodiment of the present invention, the yield is increased by least 10%. According to another embodiment of the present invention, the yield is increased by least 15%. According to another embodiment of the present invention, the yield is increased by least 30%. According to another embodiment of the present invention, the yield is increased by least 40%.

(77) Another indicator for the condition of the plant is the "plant vigor". The plant vigor becomes manifest in several aspects such as the general visual appearance. Improved plant vigor can be characterized, among others, by following improved properties of the plant: improved vitality of the plant, improved plant growth, improved plant development, improved visual appearance, improved plant stand (less plant verse/lodging), improved emergence, enhanced root growth and/or more developed root system, enhanced nodulation, in particular rhizobial nodulation, bigger leaf blade, increased plant size, increased plant weight, increased plant height, increased tiller number, increased shoot growth, increased root growth (extensive root system), increased size of root mass (extensive root system), increased yield when grown on poor soils or unfavorable climate, enhanced photosynthetic activity, change of color (e.g., enhanced pigment content (e.g., Chlorophyll content), earlier flowering, earlier fruiting, earlier and improved germination, earlier (advanced) grain maturity, improved self-defence mechanisms, less non-productive tillers, less dead basal leaves, less input needed (such as fertilizers or water), greener leaves and increased green leaf area, complete maturation under shortened vegetation periods, less fertilizers needed, less seeds needed, easier harvesting, faster and more uniform ripening, longer shelf-life, longer panicles, delay of senescence, stronger and/or more productive tillers, better extractability of ingredients, improved quality of seeds (for being seeded in the following seasons for seed production), reduced production of ethylene and/or the inhibition of its reception by the plant, spindliness of leaves, increased number of ears/m.sup.2.

(78) The improvement of the plant vigor according to the present invention particularly means that the improvement of any one or several or all of the above mentioned plant characteristics are improved independently of the pesticidal action of the composition or active ingredients. An increased vigor may for example result in a higher percentage of plants that can be transplanted to the field or an increased number of marketable plants (such as tomatoes).

(79) In certain aspects, the compositions of the present invention are applied at about 1×10⁴ to about 1×10¹⁴ colony forming units (CFU) per hectare, at about 1×10⁴ to about 1×10¹² colony forming units (CFU) per hectare, at about 1×10⁴ to about 1×10¹⁰ colony forming units (CFU) per hectare, at about 1×10⁴ to about 1×10⁸ colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10¹⁴ colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10¹² colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10¹⁰ colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10⁸ colony forming units (CFU) per hectare, at about 1×10⁸ to about 1×10¹⁴ colony forming units (CFU) per hectare, at about 1×10⁸ to about 1×10¹² colony forming units (CFU) per hectare, or at about 1×10⁸ to about 1×10¹⁰ colony forming units (CFU) per hectare.

(80) In other aspects, the compositions of the present invention are applied at about 1×10⁶ to about 1×10¹⁴ colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10¹² colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10¹⁰ colony forming units (CFU) per hectare, at about 1×10⁶ to about 1×10⁸

colony forming units (CFU) per hectare. In yet other aspects, the compositions of the present invention are applied at about 1×10^9 to about 1×10^{13} colony forming units (CFU) per hectare. In one aspect, the compositions of the present invention are applied at about 1×10^{10} to about 1×10^{12} colony forming units (CFU) per hectare.

(81) In certain embodiments, the compositions of the present invention are applied at about 0.1 kg to about 20 kg fermentation solids per hectare. In some embodiments, the compositions of the present invention are applied at about 0.1 kg to about 10 kg fermentation solids per hectare. In other embodiments, the compositions of the present invention are applied at about 0.25 kg to about 7.5 kg fermentation solids per hectare. In yet other embodiments, the compositions of the present invention are applied at about 0.5 kg to about 5 kg fermentation solids per hectare. The compositions of the present invention may also be applied at about 1 kg or about 2 kg fermentation solids per hectare.

(82) The inventive compositions, when they are well tolerated by plants, have favorable homeotherm toxicity and are well tolerated by the environment, are suitable for protecting plants and plant organs, for enhancing harvest yields, for improving the quality of the harvested material. They can preferably be used as crop protection compositions. They are active against normally sensitive and resistant species and against all or some stages of development.

(83) Plants which can be treated in accordance with the invention include the following main crop plants: maize, soya bean, alfalfa, cotton, sunflower, *Brassica* oil seeds such as *Brassica napus* (e.g., canola, rapeseed), *Brassica rapa*, *B. juncea* (e.g., (field) mustard) and *Brassica carinata*, *Arecaceae* sp. (e.g., oilpalm, coconut), rice, wheat, sugar beet, sugar cane, oats, rye, barley, millet and sorghum, triticale, flax, nuts, grapes and vine and various fruit and vegetables from various botanic taxa, e.g., *Rosaceae* sp. (e.g., pome fruits such as apples and pears, but also stone fruits such as apricots, cherries, almonds, plums and peaches, and berry fruits such as strawberries, raspberries, red and black currant and gooseberry), *Ribesioideae* sp., *Juglandaceae* sp., *Betulaceae* sp., *Anacardiaceae* sp., *Fagaceae* sp., *Moraceae* sp., *Oleaceae* sp. (e.g. olive tree), *Actinidaceae* sp., *Lauraceae* sp. (e.g., avocado, cinnamon, camphor), *Musaceae* sp. (e.g., banana trees and plantations), *Rubiaceae* sp. (e.g., coffee), *Theaceae* sp. (e.g., tea), *Sterculiaceae* sp., *Rutaceae* sp. (e.g., lemons, oranges, mandarins and grapefruit); *Solanaceae* sp. (e.g., tomatoes, potatoes, peppers, capsicum, aubergines, tobacco), *Liliaceae* sp., *Compositae* sp. (e.g., lettuce, artichokes and chicory—including root chicory, endive or common chicory), *Umbelliferae* sp. (e.g., carrots, parsley, celery and celeriac), *Cucurbitaceae* sp. (e.g., cucumbers—including gherkins, pumpkins, watermelons, calabashes and melons), *Alliaceae* sp. (e.g., leeks and onions), *Cruciferae* sp. (e.g., white cabbage, red cabbage, broccoli, cauliflower, Brussels sprouts, pak choi, kohlrabi, radishes, horseradish, cress and chinese cabbage), *Leguminosae* sp. (e.g., peanuts, peas, lentils and beans—e.g., common beans and broad beans), *Chenopodiaceae* sp. (e.g. Swiss chard, fodder beet, spinach, beetroot), *Linaceae* sp. (e.g., hemp), *Cannabaceae* sp. (e.g., cannabis), *Malvaceae* sp. (e.g., okra, cocoa), *Papaveraceae* (e.g., poppy), *Asparagaceae* (e.g., asparagus); useful plants and ornamental plants in the garden and woods including turf, lawn, grass and *Stevia rebaudiana*; and in each case genetically modified types of these plants.

(84) In certain aspects, the fermentation product further comprises a formulation ingredient. The formulation ingredient may be a wetting agent, extender, solvent, spontaneity promoter, emulsifier, dispersant, frost protectant, thickener, and/or an adjuvant. In one embodiment, the formulation ingredient is a wetting agent. In other aspects, the fermentation product is a freeze-dried powder or a spray-dried powder.

(85) Compositions of the present invention may include formulation ingredients added to compositions of the present invention to improve recovery, efficacy, or physical properties and/or to aid in processing, packaging and administration. Such formulation ingredients may be added individually or in combination.

(86) The formulation ingredients may be added to compositions comprising cells, cell-free preparations, isolated compounds, and/or metabolites to improve efficacy, stability, and physical properties, usability and/or to facilitate processing, packaging and end-use application. Such formulation ingredients may include agriculturally acceptable carriers, inerts, stabilization agents, preservatives, nutrients, or physical property modifying agents, which may be added individually or in combination. In some embodiments, the carriers may include liquid materials such as water, oil, and other organic or inorganic solvents and solid materials such as minerals, polymers, or polymer complexes derived biologically or by chemical synthesis. In some embodiments, the formulation ingredient is a binder, adjuvant, or adhesive that facilitates adherence of the composition to a plant part, such as leaves, seeds, or roots. See, for example, Taylor, A. G., et al., "Concepts and Technologies of Selected Seed Treatments," *Annu. Rev. Phytopathol.*, 28: 321-339 (1990). The stabilization agents may include anti-caking agents, anti-oxidation agents, anti-settling agents, antifoaming agents, desiccants, protectants or preservatives. The nutrients may include carbon, nitrogen, and phosphorus sources such as sugars, polysaccharides, oil, proteins, amino acids, fatty acids and phosphates. The physical property modifiers may include bulking agents, wetting agents, thickeners, pH modifiers, rheology modifiers, dispersants, adjuvants, surfactants, film-formers, hydrotropes, builders, antifreeze agents or colorants. In some embodiments, the composition comprising cells, cell-free preparation and/or metabolites produced by fermentation can be used directly with or without water as the diluent without any other formulation preparation. In a particular embodiment, a wetting agent, or a dispersant, is added to a fermentation solid, such as a freeze-dried or spray-dried powder. In some embodiments, the formulation inerts are added after concentrating fermentation broth and/or during and/or after drying. A wetting agent increases the spreading and penetrating properties, or a dispersant increases the dispersability and solubility of the active ingredient (once diluted) when it is applied to surfaces. Exemplary wetting agents are known to those of skill in the art and include sulfosuccinates and derivatives, such as MULTIWET™ MO-70R (Croda Inc., Edison, NJ); siloxanes such as BREAK-THRU® (Evonik, Germany); nonionic compounds, such as ATLOX™ 4894 (Croda Inc., Edison, NJ); alkyl polyglucosides, such as TERWET® 3001 (Huntsman International LLC, The Woodlands, Texas); C12-C14 alcohol ethoxylate, such as TERGITOL® 15-S-15 (The Dow Chemical Company, Midland, Michigan); phosphate esters, such as RHODAFAC® BG-

510 (Rhodia, Inc.); and alkyl ether carboxylates, such as EMULSOGEN™ LS (Clariant Corporation, North Carolina).
Deposit Information

(87) Samples of the *Paenibacillus* sp. strains of the invention have been deposited with the Agricultural Research Service Culture Collection located at the National Center for Agricultural Utilization Research, Agricultural Research Service, U.S. Department of Agriculture (NRRL), 1815 North University Street, Peoria, IL 61604, U.S.A., under the Budapest Treaty. *Paenibacillus* sp. NRRL B-67721, *Paenibacillus* sp. NRRL B-67723, and *Paenibacillus* sp. NRRL B-67724 were each deposited on Dec. 4, 2018. *Paenibacillus* sp. NRRL B-50374 was deposited on May 18, 2010.

(88) The *Paenibacillus* sp. strains have been deposited under conditions that assure that access to the culture will be available during the pendency of this patent application to one determined by the Commissioner of Patents and Trademarks to be entitled thereto under 37 C.F.R. § 1.14 and 35 U.S.C. § 122. However, it should be understood that the availability of a deposit does not constitute a license to practice the subject invention in derogation of patent rights granted by governmental action.

(89) The following examples are given for purely illustrative and non-limiting purposes of the present invention.

EXAMPLES

Example 1. Antifungal Activity Against *Sphaerotheca fuliginea* of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(90) *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths along with several other *Paenibacillus* sp. strains. The whole broths were diluted in a mixture of water and organic solvent to a concentration of 20%. The diluted whole broths were applied to young plants which were subsequently exposed to an inoculum of *Sphaerotheca fuliginea* (PODOXA) also known as Cucumber Powdery Mildew.

(91) Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported.

(92) All of the strains had previously been evaluated for antifungal activity in several initial rounds of screening and outperformed other *Paenibacillus* sp. strains. The present analysis represented a subsequent round of screening to further narrow the list of candidate strains with superior antifungal activity. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were among the *Paenibacillus* strains with the greatest antifungal activity in this subsequent analysis (see FIG. 1).

Example 2. Antifungal Activity Against *Botrytis cinerea* of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(93) *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in a mixture of water and organic solvent to concentrations of 20% and 5%. The diluted whole broths were applied to young plants which were subsequently exposed to an inoculum of *Botrytis cinerea* (BOTRCI) also known as Gray Mold.

(94) Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported. This assay was repeated several times and representative results are shown in Table 1. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 demonstrated consistently high antifungal activity against *Botrytis cinerea* (BOTRCI).

(95) TABLE-US-00001 TABLE 1 Control of *Botrytis cinerea* (BOTRCI) achieved with *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 at dilution rates of 20% and 5%. Application Average Percent Treatment Rate Control

<i>Paenibacillus</i> sp. strain NRRL B-50374	20%	98	5%	85
<i>Paenibacillus</i> sp. strain NRRL B-67721 <td>20%</td> <td>88</td> <td>5%</td> <td>0</td>	20%	88	5%	0
<i>Paenibacillus</i> sp. strain NRRL B-67723 <td>20%</td> <td>98</td> <td>5%</td> <td>68</td>	20%	98	5%	68
<i>Paenibacillus</i> sp. strain NRRL B-67724 <td>20%</td> <td>60</td> <td>5%</td> <td>50</td>	20%	60	5%	50

Example 3. Antifungal Activity Against *Puccinia triticina* of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(96) *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in a mixture of water and organic solvent to concentrations of 20% and 5%. The diluted whole broths were applied to young plants which were subsequently exposed to an inoculum of *Puccinia triticina* (PUCCRT) also known as Wheat Leaf Rust.

(97) Several days after exposure to the inoculum of plant pathogen, each plant was scored for percent control of the pathogen relative to the untreated control plants. Each treatment was evaluated with three replicates and the average percent control was reported. This assay was repeated several times and representative results are shown in Table 2. *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 demonstrated consistently high antifungal activity against *Botrytis cinerea* (BOTRCI).

(98) TABLE-US-00002 TABLE 2 Control of *Puccinia triticina* (PUCCRT) achieved with *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain

NRRL B-67724 at dilution rates of 2% and 5%. Application Average Percent Treatment Rate Control *Paenibacillus* sp. strain NRRL B-50374 20% 77 5% 17 *Paenibacillus* sp. strain NRRL B-67721 20% 80 5% 17 *Paenibacillus* sp. strain NRRL B-67723 20% 65 5% 25 *Paenibacillus* sp. strain NRRL B-67724 20% 80 5% 50

Example 4. Analysis of Fusaricidin and Tridecaptin Biosynthetic Gene Clusters in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(99) The genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724 were sequenced and analyzed for the presence of known gene clusters encoding proteins responsible for the biosynthesis of antimicrobial compounds. This analysis revealed that each of the four strains has a fusaricidin synthetase (*fusA*) with at least 90.5% sequence identity to that found in the other strains (see Table 3). In addition, all four strains possess a tridecaptin gene cluster. Identification of a tridecaptin gene cluster in a laboratory strain of *Paenibacillus terrae* had previously been reported. Lohans C T, van Belkum M J, Cochrane S A, Huang Z, Sit C S, McMullen L M, and Vederas J C. (2014) Biochemical, Structural, and Genetic Characterization of Tridecaptin A1, an Antagonist of *Campylobacter jejuni*. *ChemBioChem* 15:243-249. The tridecaptin biosynthetic gene cluster contains *triA*, *triB*, *triC*, *triD*, and *triE*. In each of the four strains, the gene encoding the nonribosomal peptide synthetase (NRPS) *TriE* exhibited at least 90.0% sequence identity to that found in the other strains (see Table 4). The nucleotide sequences for the *fusA* and the *triE* genes from the four strains are presented in Table 5 and Table 6, respectively. Without wishing to be bound to any theory, the expression of fusaricidins and tridecaptin in these strains may contribute to their capacity to effectively colonize plants and prevent plant disease.

(100) TABLE-US-00003 TABLE 3 Percent identity matrix created with Multiple Alignment using Fast Fourier Transform (MAFFT) for the *fusA* genes from *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724. Percent Sequence Identity NRRL NRRL NRRL NRRL Strain B-67724 B-50374 B-67721 B-67723 NRRL 100.00 97.34 97.80 90.51 B-67724 NRRL 97.34 100.00 98.18 90.71 B-50374 NRRL 97.80 98.18 100.00 90.67 B-67721 NRRL 90.51 90.71 90.67 100.00 B-67723

(101) TABLE-US-00004 TABLE 4 Percent identity matrix created with MAFFT for the *triE* genes from *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724. Percent Sequence Identity NRRL NRRL NRRL Strain B-67724 B-50374 B-67721 B-67723 NRRL 100.00 97.58 97.65 90.06 B-67724 NRRL 97.58 100.00 98.48 90.28 B-50374 NRRL 97.65 98.48 100.00 90.09 B-67721 NRRL 90.06 90.28 90.09 100.00 B-67723

(102) TABLE-US-00005 TABLE 5 Nucleotide sequences of genes encoding the fusaricidin synthetase (*fusA*) in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724. SEQ ID Strain NO: Sequence NRRL 1

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B-50374
TCAGTGTCACAGCTTTCCGGTACAGCTAAGATGAAGGGGCATATCAATATTGCTGCCTTTATGCAGTCCATTAATTTG
ATTATCAAACAGTATGATGCGTTCCGCATCCGTATTACCTCAGTGGATGGAGTGCCTCAGCAGTACGTCGTTCCCTTAT
GAAGAGAGACAGCTGGAGTGCCTGGATCTTAGCCACTATGAAAGTGTATCTGAGGTGGAAGCCTTACTTGAGCAACA
CAAAAGCAAACCTTTGCCCCTGCTGGATTCTGAGCTCTTCCAGTTTTTAATTGTGAAGATTAGCGAGGAAGAGTATTG
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CATGGAGTTAATTCAAGGAAATGAACCGAAGCTGGGCGACGATTGCTCGTATATTCAATATATTGCAGATGAGAATG
CATACGAACTTTCTGACAGATACCAAAAAGGATAAGGCTTACTGGCTGGATAAATTTTCTGATTTGCCTGAGCTTACGG
GTTGGAAGTCATATAATCCGTTATCTTTAAGCACCCACGCCGTTCCGGGAGCATTTTACCGTACCAGAAGTGCTATATC
ACGAGCTGCAAGCATTTTGCCAACAGAACAGGATTTCTTTGTTCCAGTTCTTCATGGGTGCGATGTATATCTACATAC
ACAAAATGACGAATCAGCCGGATGTGGTGATTGGCACTTCGTTCCGTAACCGGGGGAACAAAAAAGAGAAGCAAAA
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AGAAGAGCAGTCCATTTTGGCTCTTTCGGAAGGCGCTGCAGTCGATTATCCACGTGAGAAGACCATCCATGGCTTATT
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TATCCCGAGGAACGTATCCATTATATGCTGGAGGATTCAGGGGTCCGTCTGTTGCTCACCCAAAGTCATCTATGGGAG
AGCACCACTTTTGACGGAAAGCTTGTGAGTTTGGACGAAGCTACAACGTATACAGGAGATGCTTCCAATCTGGAGAG
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ATCTCTCACAGCCAAGAGCCCAGCAGCTTTCCTGGAGTTGTTGAAAGACAACCAAGTCACCATTTTAAATCAGACGCC
GACGTATTTTATCAGGTGCTACAGGAAGAGTTAACGCACTCTTCGACAGAGCTTGGCCTTAGAAAAATCATTTTTTGG
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AATTACGGAACAACGGTCCATGTACCTACAAGGAAATCACGGAACATGAGATTGAAGCGGGGAAAAGCAATATT
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GGAGTTGCCCCGCTCGGTGTCAGAGATAATTTCTTTGAAATTGGTGGTCACTCTTTGCGGGCAACAGTGCTCGTTTCA
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ATCACCTGTGGGGGTACCGGGAGAAGTCTGTCATTGCGGGCGAAGGGGTAGCGCGGGGGTATTTCAACCAGTCGGAC
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GTCACAGAGGCTTCATCTAATAACAAGCCCATTTTGTCTTAA

(103) TABLE-US-00006 TABLE 6 Nucleotide sequences of genes encoding an NRPS (triE) involved in the biosynthesis of tridecaptin in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724. SEQ ID Strain NO: Sequence NRRL 5

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B-50374

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GACATGAGGCGCTGCGTACACATTTTATTTTGGCAGAGATACGGAACCGTTGCAGGTGGTGTATCGAAATCGGGATT
GCGGCTTTC AATATGAAGATTTACACCATCTGGATGAGGATGAAATAGATACCTGGGTGAAAAATTTCAAGCTACAA
GATAAAGCACGGGGCTTTGATCTGCGCCGGGATGTCTGTCTGCGTGTAGCGATTTTACGTACTGGAGAAGACAGCTA
TCATTTTGTATGGAGCTTCCATCATATCGTCATGGACGGCTGGTGTCTCTCCCTTATAAATAAAGAAGTGTTTGAAAG
CTATGCAGCACTTCAGGAGGGCAGAGTACCAGAGCTGGCACCGGCAGTGCCGTACAGCCGCTTTATTGAATGGCTGCG
AAGCACAGGATCGCAAGGCGGCAACCGACTACTGGAGCAGCTATTTATCCGGATATGAGCAGCAAACAGCGTTACC
AGCTGTAAAATCCGGTCGCAAGAGTGAAGGCAACACGGCTAGTGATTGGGTGACGGTTTTTGGAACGTGAGTTGACCC
TCCGGGTGGAGGAGACGGCTAAGCGATATCAAGTGACCATGAATACGTTATTACAAACCGCATGGGGGATTGTGCTG
CAAAAATATAACAACACAGTGATGTCGTATTTGGCAGTGTCGTCTCAGGCCGTCCATCAGATATTATCGGGGTAGA
GGATATTATCGGCTTGTTCAATTAATACCATTCCCGTTCGCATTCTTAGTAAGGCAGGGGAATCTTTTGCAGAAGTTATG
AAGAAAACGCAGGAGCAGGCGCTGGCTTCTCATGCATATGATACGTATCCACTGTTTGAGATT CAGGCATTGACCGA
CCAGAAACAGGATTTAATTAACCATATTATGGTGTTTGAAAATTATCCGGTAGATGAGCAGGTGCAAGAAGTGGGAA
GTGCCGGGCAGGATACATTCCCGATTTCCAACGTGGTGGCTGCAGAGCAGACTA ACTATGATCTGAGCTTAGTCGTTA
TGCCGGGAGAATGCATCAAGATTCGTTTTATGTATAATGCGTTAAGTTATGATCAAACAGGCATTGAGCGTCTGCATG
GGCATTTTGTGAATCTGTTGGAGCAAGTTTTGCTTAACCCGAATGTTTGCGTAGAAGAGCTGGAACCTCGTTACAGCGG
CGAAAAAACAACAAATTACAGGAGAGTTTAATGACACTGCCTCTGCATATCCAAGCCATCACACGATCCAAAAACTG
TTTGAGGAGCAGGCAGAGCGCACGCCGGATCATATTGCTGTAGCTTTGGGT CATCAATCATTGACCTATCGGGAACCTC
AATGAGACAGCCAACCGTTTAGCGCATA CGCTACGTGATGCAGGGGTGAAATCCGATGAACCTGTGGGCATTTTGAC
GGAGCGCTCACTGGATATGATTACGGGAACACTCGCTATTTTGAAGGCTGGAGGCGCGTATGTGCCGGTAGACCCTC
AATATCCGGAGGACCGTATCCGTTATATGCTGGAGGACTCGGGAGCTAAGCTGCTGCTGGCTCAGCAGGATTTACTG
GATCGTTGCTATTTTGACGGACAGATTGTCAATCTGAATGAGGATACGTCCTACAGTGCAGATGCTTCCAATCTGGGT
ATAGATGGAGCGGGTAATCATGCTGCTTATGTCAATTATACCTCAGGTTCAACAGGTAAGCCTAAGGGTGTCGTAGTT
GAGCATCAAAGTGTCGTGCGCCTTGTCGCAATACGGATTATGTGCCATTTGATGAATCGACCCGAATGCTGCAGACT
TGCGCGTTTGATTTGATGTATCTACGTTTGAAATTTGGGGTGCGCTACTGAATGGTGGTCAGCTTGTTCTTGTGCATA
AGGATGATCTATTGGACGCGGCCAAGCTTAAAGAGACGATACGGGATCATCACGTCACCATGATGTGGCTGACCACA
CCGTTATTTAATCAGCTTTCACAGCAGGATAGTAAACTTTTCGGCGATGTGAAGTATTTGCTGGTTGGTGGTGATGTTT
TGTCTGCACCCCATATTAACCGGGTCCTGCGCGATAATCCGGACATGAACATCATTAAACGGCTATGGGCCAACGGAA
AACACAACCTTTTCGACAACGTATCACATTACGGAAGAACAGCTGGATTCTGTGCCGATCGGACGGCCCATCTGCAA
TTCGACGGCGTATGTTGTAGATTTCGTCATTTAACCTGCAACCGGTTCGGAGCTTGGGGTGAGCTGGTCGTCGGCGGAGA
CGGAGTTGCACGTGGTTATCTAAATCGTCCTGAGCTGACGGCTGAGAGATTTCTTGCTAACCCGTGGAGGGACGGAG
ATCGCCTGTACTGTACGGGAGACCTTGTTGCTGGCGTGAAGACGGCATACTGGAATACGCTGGACGGATTGATCAA
CAGGT TAAAATTCGCGGTTACCGGATTGAGCTGGGTGAGGTGGAGGCGAGGCTGGCAAGTG TACCGTCCGTGCGGGA
TAGCGTG GTTATTGCTTTGCGGGATGGAGCAGGTCAGCAGCAATTATGTGCTTACTTTACTGCCGATGAACAGCTGAC
CATCCGTGAGATTTCGAGCTGCGATGTGCGTCAGTCTGCCGAGCTATATGATTCCATCCGCATTTGTACAGCTGGATCG
GTTTCCGCTGACGACCAACGGCAAATCGACCGCAAAGCTTTACCTGTGCCGGACAAAGGGCTGCACACGGGTATAG
AATATGTGCGACCGCGAACC GATGTGGAGCAGTTGCTGGCATCTATTTGGCAGGAAGTGCTTGAAATTCCACAAGTG
GGTATCCATGATGACTTTTTTACGTTGGGCGGACATTCCCTTGAAAGTACTGGAGCTTATACGCAAAGTTCACCTTGCC
ACAGACATTGAGCTCCCCATCCGTAGTGTAATGGACTTCCCGACTATAGAAGAGCAGGCGCTCACATTATTGAAAGC
CGATCTGGAATACAAGGCCGATAGTCCAATCATTCCGTTGAATGAACACGGTCCGATCTCCATCTTCTGCTTTCCTCC
TATGCTTG GATACGGGCTGTCGTTGCGGGAGCTGGCGAAACA ACTGGATCAGGACGCAGTCGTGTATGGATTAGAGT
TCGTGGATGATGCTGCGGATGAGCAGGCCATGCTGGCACGTTACGTTGATTTGATCGTCAGCACGCAAGCGCAAGGT
CCGTATGTGCTGCTTG GTTATTCCATAGGTGGTAATCTGGCGCATAAGGTTGCTGACACACTGGAACGTCAGGGCCAT
GCTGTATCCGATATTTTATGCTCGATTCCGGTCAA AAGGGCGGAAGCCCTGCCCTTCACAGTTGAAGAAACAGAGCAT
GAAATTCATGACATGCTGGAACAGGTTCCCGAATCCTACCGTGAGCTGTTGAATGAAACGTACCAACGTAAAATGAT
CGCTTACGCGGTGTATGGCAACCAGCTTGTGAATACAGAGCGGTT CAGGCCAATATTACGGCTTTGTGCGCGGTTG
ATCGGAAACGGTGAGGGGTACAGGAGATAATCGACTTTTATGGAAAGACGCTACACAAGGTAGCTATGAAGAGCAT
AGCTTGATTGGTAATCATTATGAACTACTGGAACCCGGATTTATCGAGGAAAATGTAAAAAGTATCCGTGCCGCAAT
ACAAAAGATAACTCGGAACACGGACAAAGACATGACACAAGATTTGGCACATCATAAATCTTGA

Example 5. Characterization of Fusaricidins in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(104) Whole broth cultures of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724 were prepared by culturing each strain in a soy-based medium as in Examples 1-3. Cell extracts of the whole broth cultures were prepared, and fusaricidin-like compounds were quantified in each cell extract.

(105) A chromatographic method using ultra performance liquid chromatography/mass spectrometry triple time of flight (UPLC/MS Triple TOF) was utilized to fragment and quantify the fusaricidin-like molecules in the cell extracts: Column: YMC-Triart C8, 4.6×50 mm, 12 nm; Water (0.1% Formic Acid) and acetonitrile (0.1% Formic Acid); Gradient (% B): 0-5

min 10-70%; Wash. Mass fragmentation patterns obtained from an AB SCIEX TRIPLE TOF® mass spectrometer were analyzed to confirm the identity of each fusaricidin-like compound. The fusaricidin-like compounds analyzed included Fusaricidin A (“Fus A”), Fusaricidin B (“Fus B”), Fusaricidin C (“Fus C”), Fusaricidin D (“Fus D”), LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and LiF07b. These compounds are described in WO 2016/154297 where their structures are outlined in FIGS. 3, 4, and 12. The analytical method used to quantify the fusaricidin-like compounds did not distinguish between LiF05a and LiF06a or between LiF05b and LiF06b. Therefore, the signals produced by these two groups of compounds are referred to herein as LiF05/6a and LiF05/6b, respectively.

(106) Quantification was performed using standard curves generated with known concentrations of the fusaricidin-like compounds. Quantification of LiF05/6a, LiF05/6b, LiF07a, and LiF07b was determined from the standard curve generated with Fusaricidin C. The lowest chemical standard used was at a concentration of 0.5 µg/g. When a signal was detected from an analyte in a cell extract but the intensity of the signal was below that produced with the lowest standard, the quantity of the analyte was determined to be Below the Level of Quantification (BLOQ). The results of the quantification of the fusaricidin-like compounds in each *Paenibacillus* sp. strain are presented in Table 7.

(107) TABLE-US-00007 TABLE 7 Quantification of fusaricidin-like compounds in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724. Fusaricidins (µg/g) Strain Fus A Fus B Fus C Fus D
LiF05/6a LiF05/6b LiF07a LiF07b Total NRRL B-67724 287 78 127 31 168 32 64 8 795 NRRL B-67721 266 79 66 15 197 45 30 3 703 NRRL B-50374 167 35 53 9 217 32 23 1 536 NRRL B-67723 31 11 8 0.3 21 1 13 BLOQ 86

Example 6. Analysis of Nitrogen Fixation Gene Clusters in *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, and NRRL B-67724

(108) Additional analysis of the genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724 revealed that these strains also possess a nitrogen fixation gene cluster. The presence of a nitrogen fixation gene cluster was previously reported in the laboratory strain, *Paenibacillus* sp. WLY78. Wang L, Zhang L, Liu Z, Zhao D, Liu X, et al.

(2013) A Minimal Nitrogen Fixation Gene Cluster from *Paenibacillus* sp. WLY78 Enables Expression of Active Nitrogenase in *Escherichia coli*. *PLoS Genet* 9(10): e1003865. In *Paenibacillus* sp. WLY78, the nitrogen fixation gene cluster contains nifB, nifH, nifD, nifK, nifE, nifN, nifX, hesA, and nifV. A similar gene cluster structure was found in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724.

(109) Without wishing to be bound to any theory, the presence of the nitrogen fixation gene cluster in the genomes of *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724 may contribute to the ability of these strains to promote efficient nitrogen uptake in plants and stimulate plant growth.

(110) Comparison of the genome sequences for nifB encoding a nitrogen fixation protein showed that the nifB sequence in each strain shared at least 96.9% sequence identity with that of the other strains (see Table 8). The nucleotide sequences for the nifB sequences from the three strains are presented in Table 9.

(111) TABLE-US-00008 TABLE 8 Percent identity matrix created with MAFFT for the nifB genes from *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724. Percent Sequence Identity NRRL NRRL NRRL Strain B-67724 B-50374 B-67721 NRRL 100.00 97.27 96.93 B-67724 NRRL 97.27 100.00 99.40 B-50374 NRRL 96.93 99.40 100.00 B-67721

(112) TABLE-US-00009 TABLE 9 Nucleotide sequences of genes encoding a nitrogen fixing protein (nifB) in *Paenibacillus* sp. strains NRRL B-50374, NRRL B-67721, and NRRL B-67724. SEQ ID Strain NO: Sequence NRRL 9

ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGCATTGGAAACCCTCAGACGACATCCCTGTTATAACGAAGAG B-50374

GCACATCGCTATTTTGCGCGCATTTCATCTTCCAGTAGCCCCGGCATGCAATATTCAATGCCATTATTGCAACCGCAA
TTCGATTGCGTCAATGAAAGCCGTCCCGGCGTTGTTAGTGAAGTCTCACGCCGGAGCAGGCGGCGAGCAAGACCTA
TGCGGTAGCGGCACAGCTGATGCAGCTGTCCGTTGTCGGCATTGCGGGACCTGGAGATCCGCTGGCCAATGCGGAGG
CAACCTTCGATACCTTCCGCCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACGAATGGCCTTACCT
TGATCAGGCATATCGACAGGATTGTAGAGCTGGGTATTTTCGCATGTCACGATCACGATCAATGCTGTAGATCCAGTGG
TGGGGAGCCGCATTTATGGATGGGTCTACGATGAAGGAAAACGCTATGCAGGAGAGGAGGCCGCGCGGCTGTTGATT
GACCGCCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGCGTATTGTGCAAGGTGAAGTCCGGTGCTGATTCCCGA
AGTCAATGATGCCCATCTGCCGGAGGTAGCCAGGGTAGTCAAGGAGCACGGCGCAGTGCTGCACAACATTATGCCGC
TCATCATCGCACCTGGTAGCCGGTATGAGCAGGAAGGGATGCGGGCTCCCCGTCCCCGTCTGGTCCGGCAGCTGCAG
GAGCAATGTGTGAAGCGGGGGCTGTCATTATGCGCCATTGCCGTGAGTGCAGGGCGGATGCGATTGGACTGCTGGG
CGAGGATCGGAATCAGGATTTTACATGGGAGAACATAGCGGCTGCTCCTCCCATGGATGAAGGGGCAAGGGGCACAAT
TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAGAATGGAACGCAAGGAGGGACAATCACACCACAAACAACCGTC
AACCGGGGCTGGCTGTAGCTGCCCGTTATCGGGAGATAAGCCTGAAGCGAGCTTCACCTCAAAGCCAGTCCTAATCG
CAGTGGCCAGTTCGTGGCGGAGGGAAGGTGAATCAGCATTTCGGCCGTGCCAAGGAATTTATGATCTATGAAAGCGAC
GGGACCATCGTAAATTTTCATAGGCATTCGTAAGGTGCAATCCTACTGCCACGGGAAAGCCGATTGCAATGGGGACAA
GGTTCGAGACGATGAAGGAGATCCTTTCCATGGTGCATGACTGTGCATTGCTGCTGCTGCTCCGGCATAGGCCAAGCCCC
CAAAGAGGCATTGCAGGAAGCTGGCGTGCTGCCTATTGTATGCGGCGGGGATATTGAGGAATCGGTTCTGGAATATG
TGAAATTTCTGCGTTATATGTATCCTGTGCAGAGCAATAAAGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATCAT
TCGGATTTACCCATTAAACATTTTGGAGGCTGA NRRL 10

ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGTATTGGAAACCCTCAGACGACATCCCTGTTATAACGAAGAG B-67721

GCACATCGCTATTTGCGCGCATTCTTCCAGTAGCCCCGGCGATGCAATATTCATATGCCATTATTGCAACCCGAAAA
TTCGATTGCGTCAATGAAAGCCGTCCCGGCGTTGTTAGTGAACTGCTCACGCCGGAGCAGGCGGGCAGCAAGACCTA
TGGCGTAGCGGCACAGCTTATGCAGCTGTCCGTTGTTCGGCATTGCGGGACCTGGAGATCCGCTGGCCAATGCGGAGG
CAACCTTCGATACCTTCCGCCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACAAATGGCCTTACCT
TGATCAGGCATATTGACAGGATTGTAGAGCTGGGTATTTTCGCATGTCACGATCACGATCAATGCTGTAGATCCAGTGG
TGGGGAGCCGCATTTATGGATGGGTCTACGATGAAGGAAAACGCTATGCAGGAGAGGAGGCGCGCGGCTGTTGATT
GACCGCCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGGCGTATTGTGCAAGGTGAACTCGGTGCTGATTCCCGA
AGTCAATGATGCCCATCTGCCGGAGGTAGCCAGGGTGGTCAAGGAGCACGGTGCAGTGCTGCACAACATTATGCCGC
TCATCATCGCACCTGGTAGCCGGTATGAGCAGGAAGGGATGCGGGCTCCCCGTCCCCGTCTGGTCCGGCAGCTGCAG
GAGCAATGTGCTGAAGCGGGGGCTGTCATTATGCGCCATTGCCGTGAGTGACAGGGCGGATGCGATTGGACTGCTGGG
CGAGGATCGCAATCAGGATTTTACATGGGAGAACATAGCGGCTGCTCCTCCCATGGATGAAGGGGCAAGGACACAAT
TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAGAATGGAACGCAAGGAGGGGACAATCACACCACAAAACACCGT
AACCGGGGCTGGCTGTAGCTGCCCCGTTATCGGGAGATAAGCCTGAAGCGAGCTTCACCTCAAAGCCAGTCCTAATCG
CAGTGGCCAGTCGTGGCGGAGGGAAGGTGAATCAGCATTTCGGCCCGTGCCAAGGAATTTATGATCTATGAAAGCGAC
GGGACCATCGTAAATTTATAGGCATTTCGTAAGGTGCAATCCTACTGCCACGGGAAAGCCGATTGCAATGGGGACAA
GGTCGAGACGATGAAGGAGATCCTTTCCATGGTGCATGACTGTGCATTGCTGCTGCTGCTCGTCCGGCATAGGCGAAGCCC
CCAAAGAGGCATTGCAGGAAGCTGGCGTGCTGCCTATTGTATGCGGCGGGGATATTGAGGAATCGGTTCTGGAATAT
GTGAAATTTCTGCGTTATATGTATCCTGTGCAGAGCAATAAAGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATCA
TTCGGATTTACCCATTAAACATTTTGGAGGCTGA NRRL 11

ATGGACTCTTTAGCTGATCTCTCGGAAACCCCTTAGCATTGGACACCCTTAGACGACATCCCTGTTATAACGAAGAG
B-67724

GCACATCGCTATTTTTCGCGCATTCATCTTCCAGTAGCCCCGGCATGCAATATTCAGTGCCATTATTGCAACCCGCAAA
TTCGATTGCGTCAATGAAAGCCGTCCCGGCGTTGTTAGTGAACTGCTCACGCCGGAGCAGGCGGGCAGCAAGACCTA
TGGCGTAGCGGCACAGCTGATGCAACTGTCCGTTGTTCGGCATTGCGGGACCTGGAGATCCGCTGGCCAATGCGGAGG
CAACCTTCGATACCTTCCGCCGGGTCCGTGAGACAGTTAAGGATGTCATATTCTGTCTCAGCACGAATGGCCTTACCT
TGATCAGGCATATCGACAGGATTGTAGAGTTGGGTATTTTCGCATGTCACGATCACGATCAATGCTGTAGATCCAGTGG
TGGGGAGCCGCATTTATGGATGGGTCTACGATGAAGGAAAACGCTATGCAGGAGAGGAGGCGCGCGGCTGTTGATT
GACCGCCAGCTGGCAGGCTTGAAGATGCTGGCTTCGAGAGGGCGTATTGTGCAAGGTAACTCGGTGCTGATTCCCGA
AGTCAATGATGCCCATCTGCCAGAGGTAGCCAGGGTGGTCAAGGAGCACGGCGCGGTACTGCACAACATTATGCCGC
TCATCATCGCACCCCGGCAGTCGGTATGAGCATGAAGGGATGCGGGCCCCCGTCCCCGTCTGGTCCGGCAGCTGCAG
GAGCAATGTGCTGAGGCGGGAGCTGTCATTATGCGCCATTGCCGTGAGTGACAGGGCGGATGCGATTGGACTGCTGGG
CGAGGATCGCAATCAGGATTTTACATGGGAGAACATTGGGGGCTGCTCCTCCCATGGATGAAGAGGCAAGGGGACAAT
TTCAGAAAGAACTGGATGAGAAGGTGAGAGTGAGAATGGAACGCAAGGAGGGGACAATCGCACCACAAAAAACGT
CAACCGGGGTTGGATGTAGCTGCCCCGTTATCGGAGGATAAGCCTGAAGCAAGCTTTACCTCAAAGCCAGTCCTAATC
GCTGTGGCCAGTCGTGGCGGAGGGAAGGTGAATCAGCATTTCGGTCGTGCCAAGGAATTCATGATCTATGAAAGCGA
CGGGACTATCGTAAATTTATAGGCATTTCGTAAGGTGCAATCCTACTGCCATGGGAAAGCCGATTGCAATGGAGACA
AGGTTCGAGACGATGAAGGAGATCCTTTCCATGGTGCATGACTGTGCATTGCTGCTGCTGCTCGTCCGGCATAGGCGAAGCC
CCCAAAGAGGCATTGCAGGATGCTGGCGTGCTGCCTATTGTATGCGGCGGGGATATTGAGGAATCCGTTCTGGAATA
TGTAATTTCTGCGTTATATGTATCCTGTGCAGAGCAATAAGGGAAGTAAGCGCAATAAGGGAGTTAAGGGCAATC
ATTCGGATTTACCCATTGAACATTTTGGAGGCTGA

Example 7. Plant Growth Promotion Properties of *Paenibacillus* sp. Strains NRRL B-50374, NRRL B-67721, NRRL B-67723, and NRRL B-67724

(113) *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, and *Paenibacillus* sp. strain NRRL B-67724 were cultured in a soy-based medium to produce whole broths that were diluted in water to a concentrations of 20%. The diluted whole broths were applied to tomato plants as a drench at the time of planting. Several days later when the tomato seedlings had developed into young plants the average weights and standard deviations for forty whole tomato plants in each treatment group were measured. An untreated control group (“UTC”) and a group treated with the soy-based medium only (“Medium”) were included in the analysis for comparison.

(114) Interestingly, *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, and *Paenibacillus* sp. strain NRRL B-67724, each of which possesses a nitrogen fixation gene cluster, produced a significant increase in tomato plant growth compared to the UTC and Medium controls (see FIG. 2). In contrast, *Paenibacillus* sp. strain NRRL B-67723, which does not possess a nitrogen fixation gene cluster, did not increase tomato plant growth compared to the UTC and Medium controls.

(115) Unless defined otherwise, all technical and scientific terms herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All publications, patents, and patent publications cited are incorporated by reference herein in their entirety for all purposes.

(116) It is understood that the disclosed invention is not limited to the particular methodology, protocols and materials described as these can vary. It is also understood that the terminology used herein is for the purposes of describing particular embodiments only and is not intended to limit the scope of the present invention which will be limited only by the appended claims.

(117) Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many

equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A composition comprising a preservative and a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^{10} colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.
2. The composition according to claim 1, comprising Fusaricidin A, Fusaricidin B, Fusaricidin C, Fusaricidin D, LiF05a, LiF05b, LiF06a, LiF06b, LiF07a, and/or LiF07b.
3. The composition according to claim 1, wherein the composition effectively controls a plant disease caused by a fungus selected from the group consisting of *Botrytis cinerea*, *Sphaerotheca fuliginea*, and *Puccinia triticina*.
4. The composition according to claim 1 comprising a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.
5. The composition according to claim 1, wherein the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, or *Paenibacillus* sp. strain NRRL B-67724.
6. The composition according to claim 1 wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof and further comprises a nitrogen fixation gene cluster; and the nitrogen fixation gene cluster comprises a nitrogen fixation gene of nifB encoded by a DNA sequence exhibiting at least 96.9% sequence identity to SEQ ID NO: 10.
7. The composition according to claim 6, wherein expression of the nitrogen fixation gene cluster contributes to enhanced plant growth, plant vigor, and/or crop yield.
8. The composition according to claim 6, wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 97.3% sequence identity to SEQ ID NO: 1 and the *Paenibacillus* sp. strain comprises a NRPS gene of triE encoded by a DNA sequence exhibiting at least 97.5% sequence identity to SEQ ID NO: 5.
9. The composition according to claim 6 comprising a fermentation product of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.
10. The composition according to claim 6, wherein the fungicidal mutant strain has a genomic sequence with greater than about 90% sequence identity to *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724 and/or the fungicidal mutant strain has fungicidal activity and/or levels of a fusaricidin that are comparable to or better than that of *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, or *Paenibacillus* sp. strain NRRL B-67724.
11. A method of treating an agricultural plant to control a disease, wherein the method comprises applying to a part of the plant and/or to a locus of the plant an effective amount of a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the *Paenibacillus* sp. strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^{10} colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.
12. The method according to claim 11, wherein the method comprises applying the composition to foliar plant parts.
13. The method according to claim 11, wherein the composition is applied at about 1×10^{10} to about 1×10^{12} colony forming units (CFU) of the *Paenibacillus* sp. strain per hectare or at about 0.5 kg to about 5 kg fermentation solids per hectare.
14. The method according to claim 11, wherein the disease is a fungal disease or a bacterial disease.
15. A method for increasing the vigor and/or crop yield of an agricultural plant, wherein the plant, the plant propagule, the seed of the plant and/or a locus where the plant is growing or is intended to grow is treated with an effective amount of a composition comprising a biologically pure culture of a *Paenibacillus* sp. strain or a cell-free preparation thereof comprising a fusaricidin and a tridecaptin; wherein the *Paenibacillus* sp. strain comprises a fusaricidin synthetase gene of fusA encoded by a DNA sequence exhibiting at least 90.5% sequence identity to SEQ ID NO: 3; and the *Paenibacillus* sp.

strain comprises a nonribosomal peptide synthetase (NRPS) gene of triE encoded by a DNA sequence exhibiting at least 90.0% sequence identity to SEQ ID NO: 7, wherein the culture comprises at least 1×10^{10} colony forming units (CFU) of the strain per mL and wherein the *Paenibacillus* sp. strain is *Paenibacillus* sp. strain NRRL B-50374, *Paenibacillus* sp. strain NRRL B-67721, *Paenibacillus* sp. strain NRRL B-67723, *Paenibacillus* sp. strain NRRL B-67724, or a fungicidal mutant strain thereof.

16. The method according to claim 15, wherein the treatment is carried out as an in-furrow treatment, seed treatment, and/or foliar treatment.

17. The method according to claim 11, wherein the agricultural plant is selected from the group consisting of soybean, corn, wheat, triticale, barley, oat, rye, rape, millet, rice, sunflower, cotton, sugar beet, pome fruit, stone fruit, citrus, banana, strawberry, blueberry, almond, grape, mango, papaya, peanut, potato, tomato, pepper, cucurbit, cucumber, melon, watermelon, garlic, onion, broccoli, carrot, cabbage, bean, dry bean, canola, pea, lentil, alfalfa, trefoil, clover, flax, elephant grass, grass, lettuce, sugarcane, tea, tobacco and coffee; each in its natural or genetically modified form.

18. A seed treated with the composition according to claim 1.
